

Instructions to the Students:

1. All the questions are compulsory
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question
3. Use of non-programmable scientific calculators is allowed
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

Q.1 Attempt Any Two

A) Describe the stress-strain curve for a typical ductile material. Include the major regions of the curve and explain the behavior of the material in each region. (CO1) 6

B) A composite bar made of copper, steel and brass is rigidly attached to the end supports as shown in figure 01. Determine the stresses in the three portions of the bar when the temperature of the composite system is raised by 70°C when

i) The supports are rigid ii) the supports yield by 0.6 mm.

$$E_c = 100 \text{ GPa}, E_s = 205 \text{ GPa}, E_b = 95 \text{ GPa}$$

$$\alpha_c = 18 \times 10^{-6} / ^{\circ}\text{C}, \alpha_s = 11 \times 10^{-6} / ^{\circ}\text{C},$$

$$\alpha_b = 19 \times 10^{-6} / ^{\circ}\text{C}$$

(CO2) 6

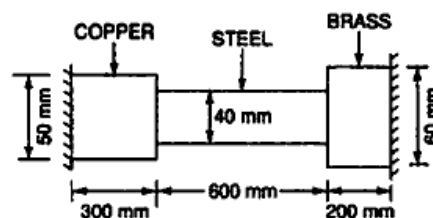


Figure 01

C) From the state of stress at a point in a strained material, it is found that $\sigma_x=200\text{MPa}$ (Tensile); $\sigma_y=150\text{MPa}$ (Compressive) and $\tau_{xy}=120\text{MPa}$. Determine analytically,

(CO3) 6

- (i) Principal stresses and direction of principal planes
- (ii) Maximum shear stress
- (iii) Normal stress on plane of maximum shear stress.

Q.2 Attempt Any Two

- | | | | |
|----|---|-------|---|
| A) | A wagon weighing 20 kN is attached to a wire rope and is moving at the speed of 5.4 kmph. The rope suddenly jams and wagon is brought to rest. If length of rope is 50 m and diameter is 36 mm, find maximum instantaneous stress and elongation of rope. Take $E = 200 \text{ GPa}$. | (CO2) | 6 |
| B) | Define short and long columns and discuss their behavior under compressive loads. Explain Euler's and Rankine's formulae for calculating the critical buckling load of a column. | (CO3) | 6 |
| C) | A steel beam with a cross-sectional area of 3000 mm^2 and a length of 4 meters is subjected to an axial load of 20 kN and a bending moment of $40 \text{ kN}\cdot\text{m}$. Calculate the maximum permissible axial load and bending moment if the middle third rule is to be satisfied. | (CO4) | 6 |

Q.3 Attempt Any Two

- | | | | |
|----|---|-------|---|
| A) | An I-section beam 350 mm X 150 mm has a web thickness 10 mm & flange thickness 20 mm. If the shear force acting on the section is 40 kN, then find; | | |
| | a) Maximum shear stress developed in the section | (CO3) | 6 |
| | b) Sketch the shear stress distribution diagram | | |
| | c) Total shear force carried by web | | |
| B) | Derive an expression for Flexural equation along with assumptions | (CO3) | 6 |
| C) | A Hollow shaft with diameter ratio of $3/8$ is required to transmit 500 kW at 100 rpm, the maximum torque being 20% greater than mean. The maximum shear stress is not to exceed 60 N/mm^2 and the twist in the length of 3 m is not to exceed 1.4° . Calculate the minimum diameter required for the shaft. | (CO4) | 6 |

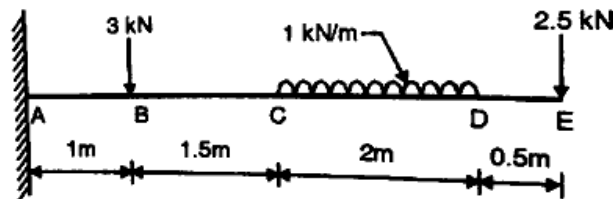
Take $G = 84 \text{ N/mm}^2$.

Q.4 Attempt Any Two

- | | | | |
|----|--|-------|---|
| A) | Draw shear force diagram (SFD) and Bending moment diagram (BMD) for a simply supported beam of length $L = 8\text{m}$ carrying vertically downward point loads of 4kN, 10kN, and 7kN at a distance of 1.5m, 4m, and 6m respectively from left hand support. Also determine the maximum bending moment developed in the | (CO5) | 6 |
|----|--|-------|---|

beam.

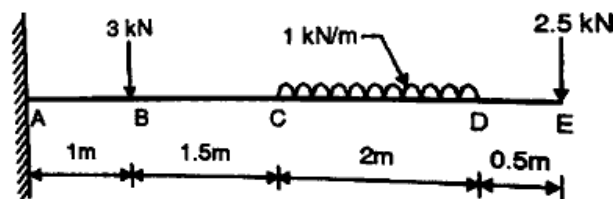
- B) Draw shear force and bending moment diagrams for the cantilever beam as shown in figure 02



(CO5) 6

Figure 02

- C) Draw shear force and bending moment diagrams for the cantilever beam as shown in figure 03



(CO5) 6

Figure 03

Q. 5 Attempt Any Two

- A) Explain the principle of Macaulay's method and describe the steps involved in applying Macaulay's method for calculating deflection in beams subjected to complex loading conditions.

(CO6) 6

- B) A composite beam consists of a steel section and a timber section, each 2 meters long. The beam is simply supported with a span of 4 meters and subjected to a triangular load varying from 0 kN at one end to 20 kN at the other end. Determine the total deflection at the midpoint of the beam using the moment-area method. Given: $E_{\text{steel}}=200 \text{ GPa}$, $E_{\text{timber}}=12 \text{ GPa}$, $I_{\text{steel}}=400 \text{ mm}^4$, $I_{\text{timber}}=150 \text{ mm}^4$.

(CO6) 6

- C) An overhanging beam is supported at both ends and has two unequal spans of lengths 3 meters and 4 meters, respectively. The beam is subjected to a uniformly distributed load of 15 kN/m over the entire length and a point load of 40 kN at a distance of 1 meter from the left support. Calculate the maximum deflection at the midpoint of the longer span using the double integration method. Given: $E=210 \text{ GPa}$, $I=600 \text{ mm}^4$.

(CO6) 6

END